⚠ EARLY-VERSION TEST MOCK-UP, ILLUSTRATES THE IDEA AND THE GOAL ONLY · NOT A DESIGN · NOT FOR CONSTRUCTION · EVERYTHING SUBJECT TO COMPLETE CHANGE ⚠

*VERY EARLY CONCEPT ILLUSTRATION, NOT A DESIGN DOCUMENT. The SELENE design map, companion to the BRONTE Chimborazo sheet (KER-BRO-TRK-001). Two machines, one ridge: **EOS-1** (Ø1.2 m pod, built first, cargo return to Earth) and **EOS-2** (Ø2.0 m pod, second machine, second pad, bigger cargo, future). The site map is rendered from real NASA LRO/LOLA laser topography (20 m/px polar DEM); only the track alignment drawn on it is conceptual. All physics figures re-derived June 12, 2026.*

ANNEX A

SELENE, South Pole Design Map

KER-SEL-TRK-002 · REV D · EARLY-VERSION TEST MOCK-UP · NOT FOR CONSTRUCTION

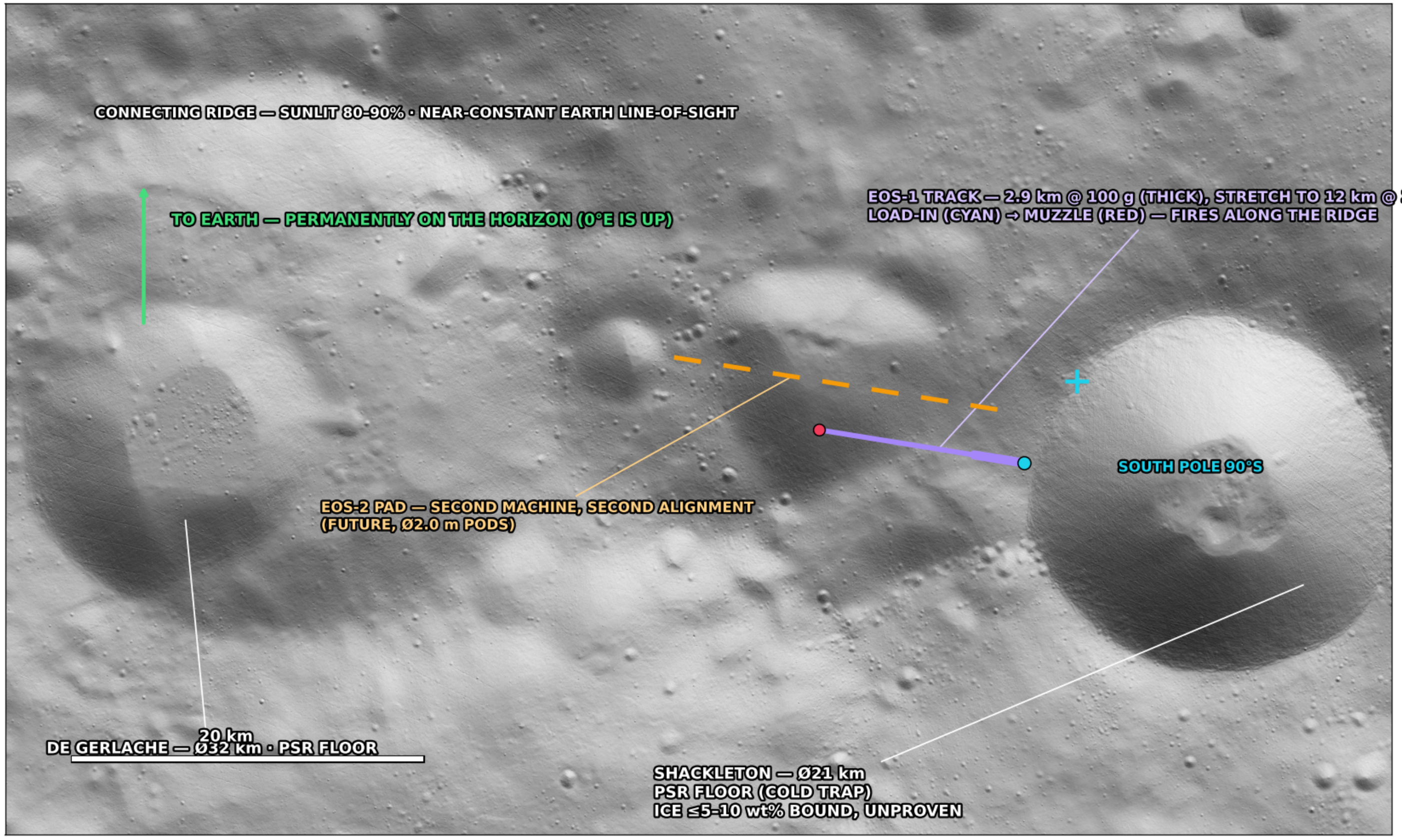
WHAT IS INSIDE

- Panel A, Site: real LRO/LOLA laser topography, Shackleton / Connecting Ridge ~89.7°S
- Panel A2, Where on the Moon · what does not exist here · two machines (EOS-1 / EOS-2)
- Panel B, EOS-1 open-track section: no tube, the Moon is the vacuum vessel
- Panel C, Side profile: the raised track on its regolith embankment (true 1:1)
- Charts, track-length vs g, Δv destinations, ridge illumination, engineering

> *Each large panel is on its own page, rotated to read at full size, turn the page (or your screen) a quarter-turn clockwise.*

PANEL A, SITE: REAL LRO/LOLA TOPOGRAPHY, SHACKLETON / CONNECTING RIDGE, ~89.7°S

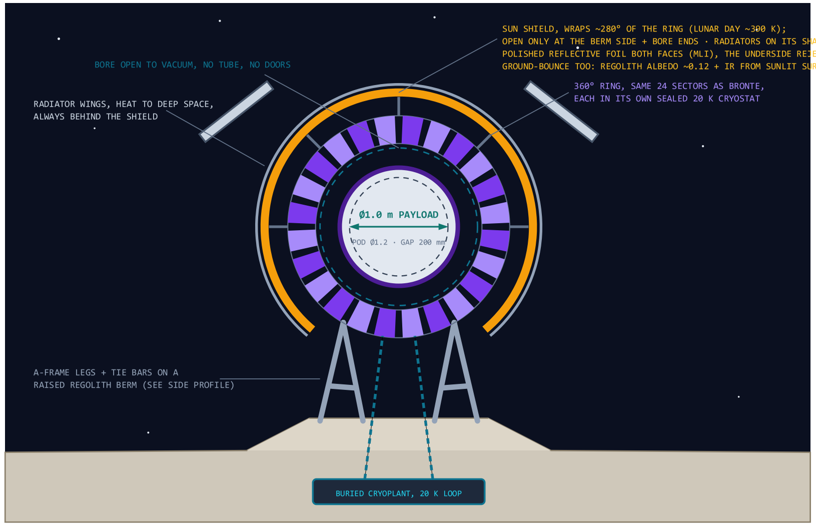
LUNAR SOUTH POLE — REAL LRO/LOLA TOPOGRAPHY (20 m/px, NASA GSFC) · HILLSHADE RENDER · EARLY CONCEPT, NOT FOR CONSTRUCTION



PANEL B, EOS-1 OPEN-TRACK SECTION, NO TUBE: THE MOON IS THE VACUUM VESSEL

EOS-1 SECTION, THE SELENE RING, OPEN TO VACUUM, Ø1.6 m BORE · POD Ø1.2 m · GAP 200 mm

DRAWN TO SCALE (1 m ≈ 110 px) · 24-SECTOR REBCO RING @ 20 K IN SEALED CRYOSTATS · WRAP-AROUND SUN SHIELD COVERS ~280° OF THE RING



REGOLITH, ~250 K CONSTANT BELOW 2 m: NATURAL COLD BUFFER FOR THE CRYO LOOP

POD "EOS-1", DAWN, SISTER OF SELENE, MISSION ONE: CARGO HOME

FULL-HULL REACTION SLEEVE, SAME 360° COUPLING, GAP 200 mm

Ø1.2 m × 7 m · 500 kg GROSS · REAR ENGINE Δv 0.77 km/s (Isp 350) · RCS QUADS, NO FINS

PAYLOAD BAY Ø1.0 m × 4 m, RACKS ARE INTERCHANGEABLE: CARGO-RETURN CANISTERS FIRST · SATELLITE DISPENSERS LATER · SAME POD, SAME RING

MISSION ONE: ESCAPE @ 2.38 km/s → REAR-ENGINE TRIMS → COAST SHIELD-FORWARD → DIRECT SHIELD-FIRST ENTRY → CHUTES → RECOVERY. THE POD IS THE SHIPPING CRATE.

HARDENED TO 100 g, ARTILLERY ELECTRONICS QUALIFY TO 15,000–20,000 g

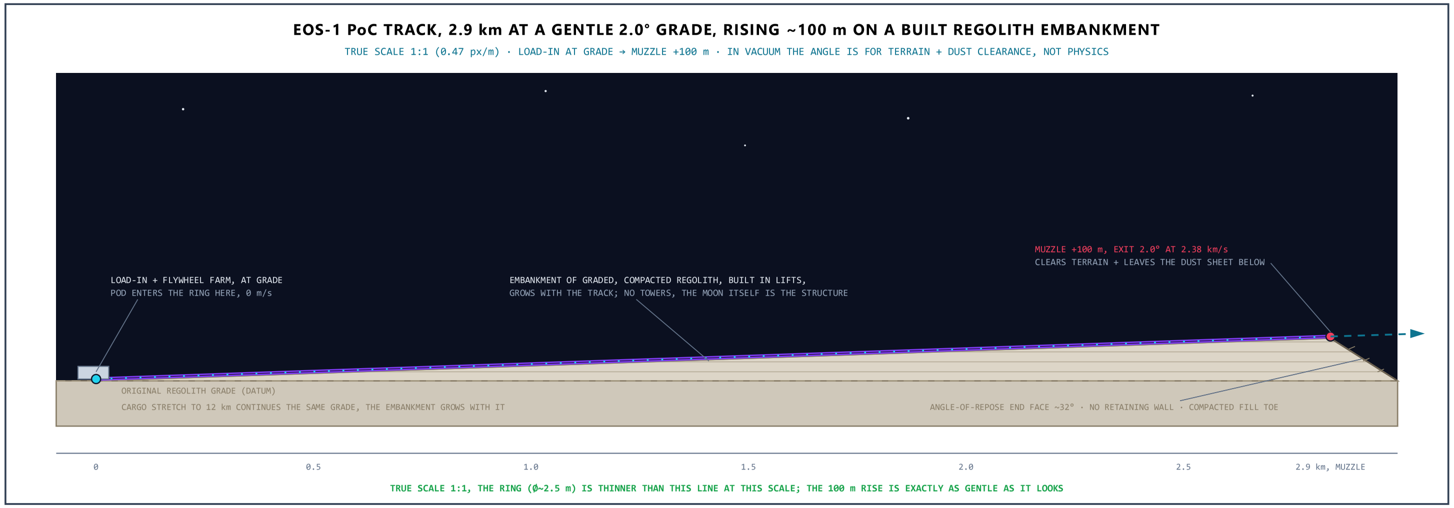
ONE BLUNT FACE SERVES BOTH WORLDS: DUST AT LAUNCH, PLASMA AT EARTH.

THE HONEST HARD PARTS (LUNAR EDITION)

1. DUST, ELECTROSTATIC + EJECTA. SHIELD WRAP, PLINTHS, AND THE DOME EXIST BECAUSE OF IT. APOLLO'S #1 MECHANICAL COMPLAINT.
2. THERMAL SWINGS, 390 K SUN TO CRYO-COLD SHADOW ON ONE TRACK. ANSWERED BY THE 280° SHIELD + SEALED CRYOSTATS + BURIED LOOP.
3. THE PULSE, 0.39 MWh IN 2.4 s = 1.17 GW FROM FLYWHEELS CHARGED BY ~10 MW OF RIDGE SOLAR. STORAGE MASS RIDES THE FIRST ROCKETS.
4. CATCHING / RECOVERY, MISSION ONE SIDESTEPS IT: EARTH ITSELF IS THE CATCHER'S MITT (SHIELD + CHUTES). CISLUNAR CATCH COMES LATER.
5. ICE UNCERTAINTY, SHACKLETON BOUNDED ≤5–10 wt%, UNPROVEN. THE PLAN CLOSES ON REGOLITH/O_2/METAL IN THE LEAN CASE.
6. NO IN-FLIGHT ABORT, A THROWN POD IS COMMITTED. RANGE SAFETY IS TRAJECTORY DESIGN, DECIDED BEFORE THE SHOT.

MISSION FROM THIS LIST IS THE POINT: NO WINDOW, NO TUBE, NO BOOM, NO LAUNCH HEATING, NO WEATHER, NO RESERVE LAWYERS.

PANEL C, SIDE PROFILE: THE RAISED TRACK ON ITS EMBANKMENT (TRUE SCALE 1:1)



PANEL A2, WHERE ON THE MOON · WHAT DOES NOT EXIST HERE · TWO MACHINES

WHERE ON THE MOON?

HERE, SOUTH POLE (BOTTOM EDGE, AS SEEN FROM EARTH)

PHOTO: REAL NEARSIDE (FULL MOON) · PANEL A MAP: REAL LRO/LOLA LASER TOPOGRAPHY

WHAT DOES NOT EXIST ON THIS SHEET

NO VACUUM TUBE, THE TRACK IS ALREADY IN HARD VACUUM

NO PLASMA WINDOW, NO ATMOSPHERE TO KEEP OUT

NO AERO-SHELL, NO NEEDLE NOSE, NOTHING TO CUT THROUGH

NO ISOLATION GATES, NO BLAST SHUTTERS, NO TUBE TO BREACH

NO SONIC BOOM, NO LAUNCH HEATING, NO DRAG, EVER

EVERY HARD SUBSYSTEM ON THE BRONTE SHEET IS DELETED BY ONE SENTENCE: THE MOON HAS NO ATMOSPHERE. WHAT REMAINS IS THE RING, THE POWER, THE COLD, AND THE MOON SUPPLIES THE COLD.

WHAT REPLACES THEM: DUST, 250–390 K THERMAL SWINGS, GW-CLASS PULSE STORAGE, AND CATCHING THE CARGO ON THE OTHER END.

SITE SERVICES: RIDGE SOLAR 80–90% UPTIME · PSR COLD SINK NEXT DOOR · REGOLITH ~250 K AT 2 m DEPTH, NATURAL CRYO BUFFER

MAP DATA: NASA LRO/LOLA POLAR DEM, 20 m/px (LDEM_875S_20M), POLAR STEREOGRAPHIC, RENDERED AS HILLSHADE, A REAL SURVEY, UNLIKE EVERYTHING ELSE ON THIS CONCEPT SHEET.

TWO MACHINES, ONE RIDGE, BUILD SMALL, THEN WIDE

EOS-1, FIRST BUILD · POD Ø1.2 m · BORE Ø1.6 m · GAP 200 mm
500 kg GROSS · CARGO RETURN TO EARTH IS MISSION ONE
0.39 MWh/SHOT · PEAK 1.17 GW · RECHARGE 2.4 min @ 10 MW
~25 SHOTS/HR → ~300 t/DAY THROWN FROM ONE STARTER TRACK

EOS-2, SECOND MACHINE · POD Ø2.0 m · BORE Ø2.4 m
~2.5 t GROSS · BIGGER UNIT CARGO + SATELLITE DISPENSERS
1.96 MWh/SHOT · PEAK 5.8 GW · ~5 SHOTS/HR @ 10 MW
SECOND PAD, SAME RIDGE, BUILT AFTER EOS-1 PROVES THE KIT

SAME ~300 t/DAY EITHER WAY, THROUGHPUT IS SET BY THE SOLAR PLANT, NOT THE POD. EOS-2 BUYS SIZE, NOT MASS.

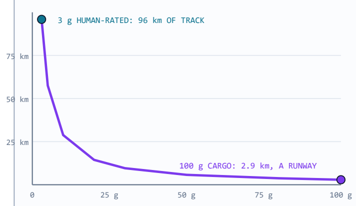
LADDER: 2.9 km @ 100 g → 12 km @ 30 g → ~50 km @ 3 g
ENERGY: 0.41 ORBIT · 0.78 ESCAPE · 1.25 MARS XFER kWh/kg
PELLET MODE (SJSU 2023): 25.4 kg EVERY 10 s ON 8.7 MW

FIRST BILL IS A ROCKET BILL: STARTER COILS + PLANT MUST FIT 1–2 HEAVY-LIFT FLIGHTS

LAUNCH AZIMUTH ALONG THE RIDGE; ORBIT PLANE PICKED BY LAUNCH TIME, A POLAR SITE COVERS ALL GEOMETRY MONTHLY

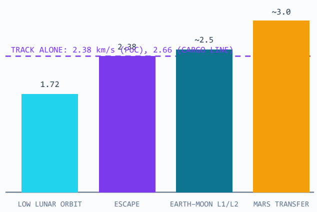
DETAILS, SITE & SYSTEM CHARTS

TRACK LENGTH FOR ESCAPE vs ACCELERATION



L = $v^2/2a$ AT v = 2.38 km/s. ACCELERATION TOLERANCE IS THE WHOLE COST MODEL: HARDENED CARGO MAKES THE STARTER TRACK TINY.

WHERE THE TRACK + KICK GETS YOU (Δv FROM SURFACE)



THE 0.77 km/s KICK STAGE TOPS UP THE REST: EVERYTHING IN CISLUNAR SPACE, AND A MARS SHIPMENT, IS IN REACH.

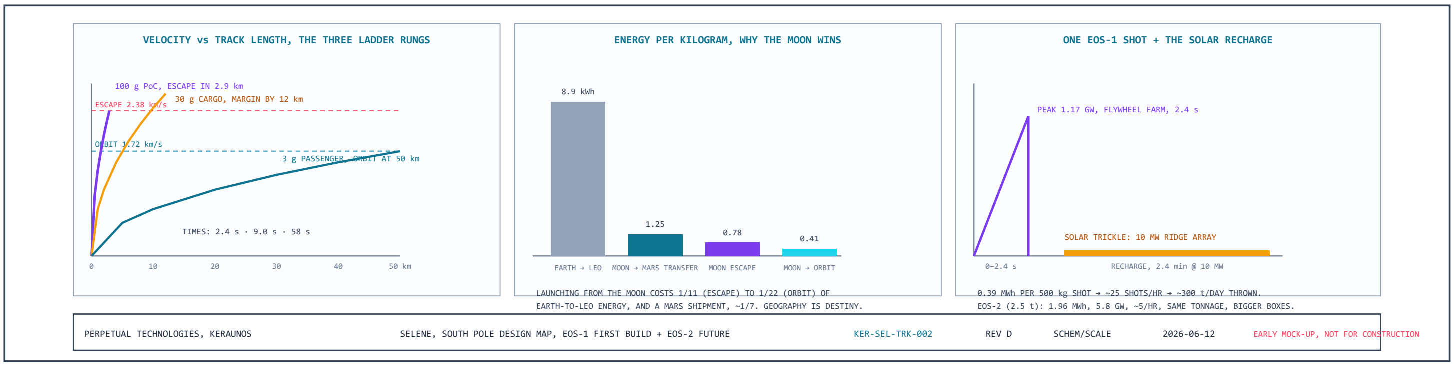
RIDGE ILLUMINATION OVER A YEAR (LITERATURE, SCHEMATIC)

GOLD = SUNLIT (80–90% OF THE YEAR AT THE BEST RIDGE SITES)

DARK SPELLS RUN HOURS TO A FEW EARTH-DAYS (LONGEST KNOWN OUTAGES ~3–5 DAYS), BRIDGED BY THE SAME FLYWHEEL FARM THAT FIRES THE TRACK; LAUNCHES PAUSE, NOTHING FREEZES.

WHY NO MONTHS-LONG EXTREMES: LUNAR AXIAL TILT IS ONLY 1.5°, THE MOON BARELY HAS SEASONS. THE ±130 °C SWINGS ARE THE EQUATOR'S 14-DAY NIGHT PROBLEM; POLAR RIDGES ARE THE MOST THERMALLY STABLE GROUND ON THE MOON. (TIMELINE SCHEMATIC.)

DETAILS, SELENE ENGINEERING CHARTS



REV D, pod architecture per direction: the PICA heat shield moves to the FRONT, where it pulls double duty as the launch dust dome (one blunt face serves both worlds, dust at launch, plasma at Earth), and a correction engine sits at the REAR for midcourse and Earth-targeting burns; the pod flies shield-first the whole trip. The section is now titled the SELENE ring (parts-shared with BRONTE, but its own machine). The ground geometry is rebuilt, one continuous regolith polygon with a flat-topped berm the legs actually stand on, and a new Panel C shows the BRONTE-style side profile at true 1:1: the 2.9 km track rising ~100 m at 2.0° on a growing regolith embankment built up in compacted lifts and ending in an angle-of-repose (~32°) face rather than a cliff, load-in at grade, muzzle elevated to clear terrain and the dust sheet, with the honest vacuum note that the angle is for clearance, not physics. Prior note: the Moon is now real: Panel A is a hillshade render of NASA LRO/LOLA laser topography (20 m/px polar DEM; projection verified by landing the test markers inside Shackleton itself), and the locator inset uses a real nearside photograph. The sun shield gained its material spec, polished reflective foil on both faces, because the underside sees regolith ground-bounce (albedo ~0.12) and surface IR even though the sun never gets beneath it. A new charts row covers what the old sheet only asserted: track-length-versus-g (why hardened cargo makes the starter track runway-sized), the Δv bars showing track-plus-kick reaching everything cislunar plus a Mars shipment, and the ridge illumination year with its few-day dark spells bridged by the same flywheels that fire the track. Prior note: geometry regenerated clean (24 even sectors, symmetric A-frames with tie bars; no more wonk), the sun shield now wraps ~280° of the ring, a gold thermal shell open only toward the berm and the bore ends, with the radiator wings mounted on its permanently shaded back, and Panel A opens with a "where on the Moon?" locator: this is the very bottom of the Moon, ~89.7°S, the Shackleton–de Gerlache Connecting Ridge. The pod line is now two machines: ***EOS-1 first****, Ø1.2 m × 7 m, 500 kg, bore Ø1.6 m at the standard 200 mm gap, whose payload bay takes interchangeable racks (cargo-return canisters first, satellite dispensers later) and whose mission one is shipping lunar material home: escape at 2.38 km/s, kick trim, shield-first reentry, chutes, the pod is the shipping crate, and the heat shield earns its mass only at Earth.* ***EOS-2 follows****, Ø2.0 m, ~2.5 t, its own pad on the same ridge, and the charts carry the quiet insight: at a fixed 10 MW solar plant both machines throw the same ~300 tonnes a day; the bigger ring buys bigger boxes, not more mass. Throughput belongs to the power plant.*